CHAPTER 2 - Introduction to LP

Ex. 2.1-1 The Reddy Mikks Company

Produces interior and exterior paints from raw materials M1 and M2

	Exterior	Interior	Daily Availability
M1	6	4	24
M2	1	2	6
Profit per ton	5	4	

- Daily demand for interior paint cannot exceed that of exterior by more than 1 ton.
- Maximum demand for interior is 2 tons.
- What is the optimum product mix of exterior and interior which maximizes the total daily profit?

All OR models, LP included, consist of three basic components:

- Decision variables?
- Objective?
- Constraints?

Let, = amount of exterior paint produced daily

= amount of interior paint produced daily

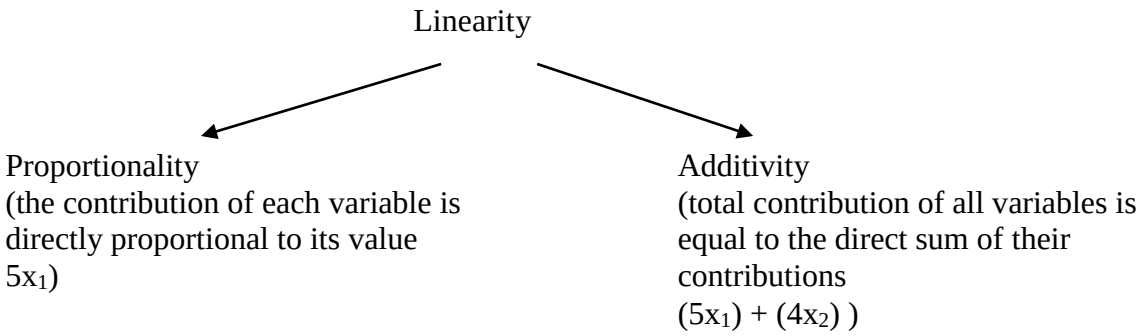
The goal is …

s.t.

- Any values of x_1 and x_2 satisfying all constraints is a feasible solution.

- The best among all feasible solutions is called the optimum solution.

In LP, the objective and constraints functions are linear.



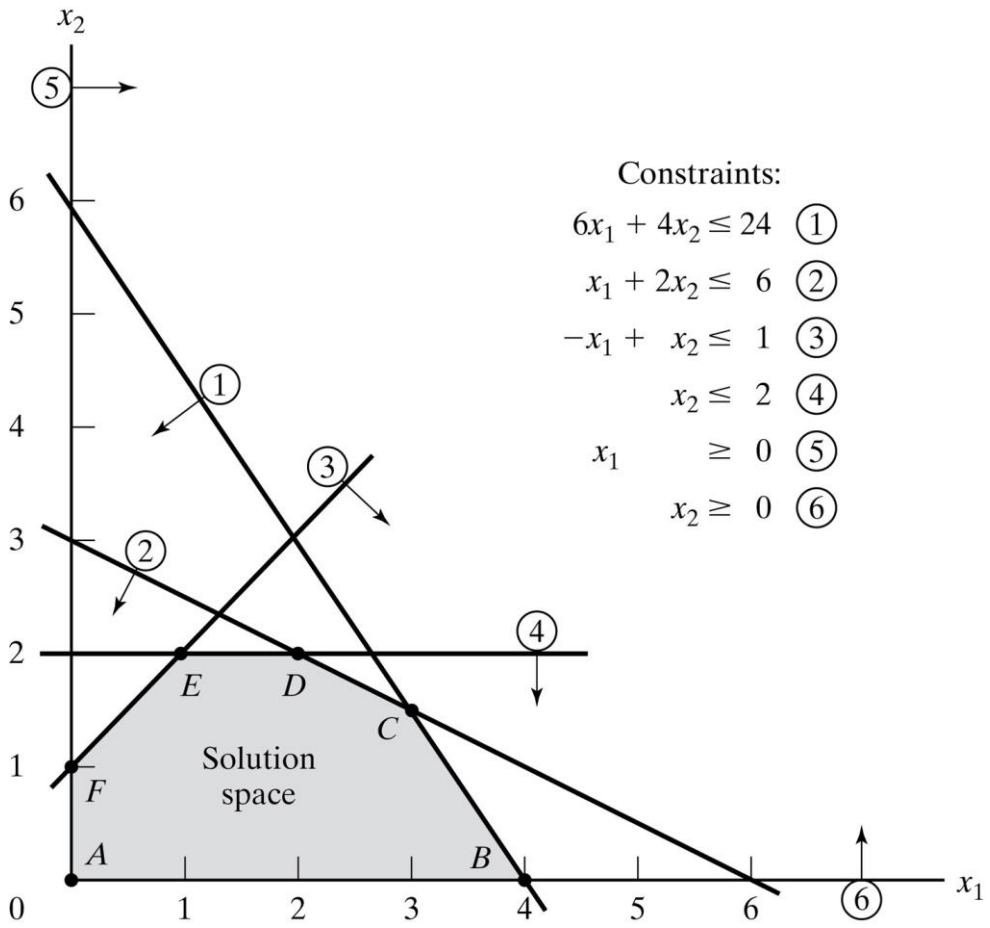
Graphical Solution

$6x_1 + 4x_2 = 24 \quad \Rightarrow \quad x_1 = 0 \;\Rightarrow\; x_2 = 6\,, \qquad x_2 = 0 \;\Rightarrow\; x_1 = 4$

$x_1 + 2x_2 = 6 \quad \Rightarrow \quad x_1 = 0 \;\Rightarrow\; x_2 = 3\,, \qquad x_2 = 0 \;\Rightarrow\; x_1 = 6$

$-x_1 + x_2 = 1 \quad \Rightarrow \quad x_1 = 0 \;\Rightarrow\; x_2 = 1\,, \qquad x_2 = 0 \Rightarrow\; x_1 = -1$

$x_2 = 2$



$5x_1 + 4x_2 = 10$ $\quad x_1 = 0 \quad x_2 = 0$

$\quad x_2 = 2.5 \quad x_1 = 2$

$5x_1 + 4x_2 = 15$ $\quad x_1 = 0 \quad x_2 = 0$

$\quad x_2 = 3.75 \quad x_1 = 3$

$6x_1 + 4x_2 = 24$

$x_1 + 2x_2 = 6$

$4x_1 = 12 \Rightarrow x_1 = 3$

$x_2 = 1.5$

$z = 5x_1 + 4x_2 = 5 \times 3 + 4 \times 1.5 = 21$

Ex. 2.2–2 Diet Problem

Ozark farms uses at least 800lb of special feed daily (mixture of corn and soybean)

Feedstuff	Protein	Fiber	Cost ($/lb)
Corn	0.09%	0.02%	0.30
Soybean	0.60%	0.06%	0.90

Dietary requirements: at least 30% protein
at most 5% fiber

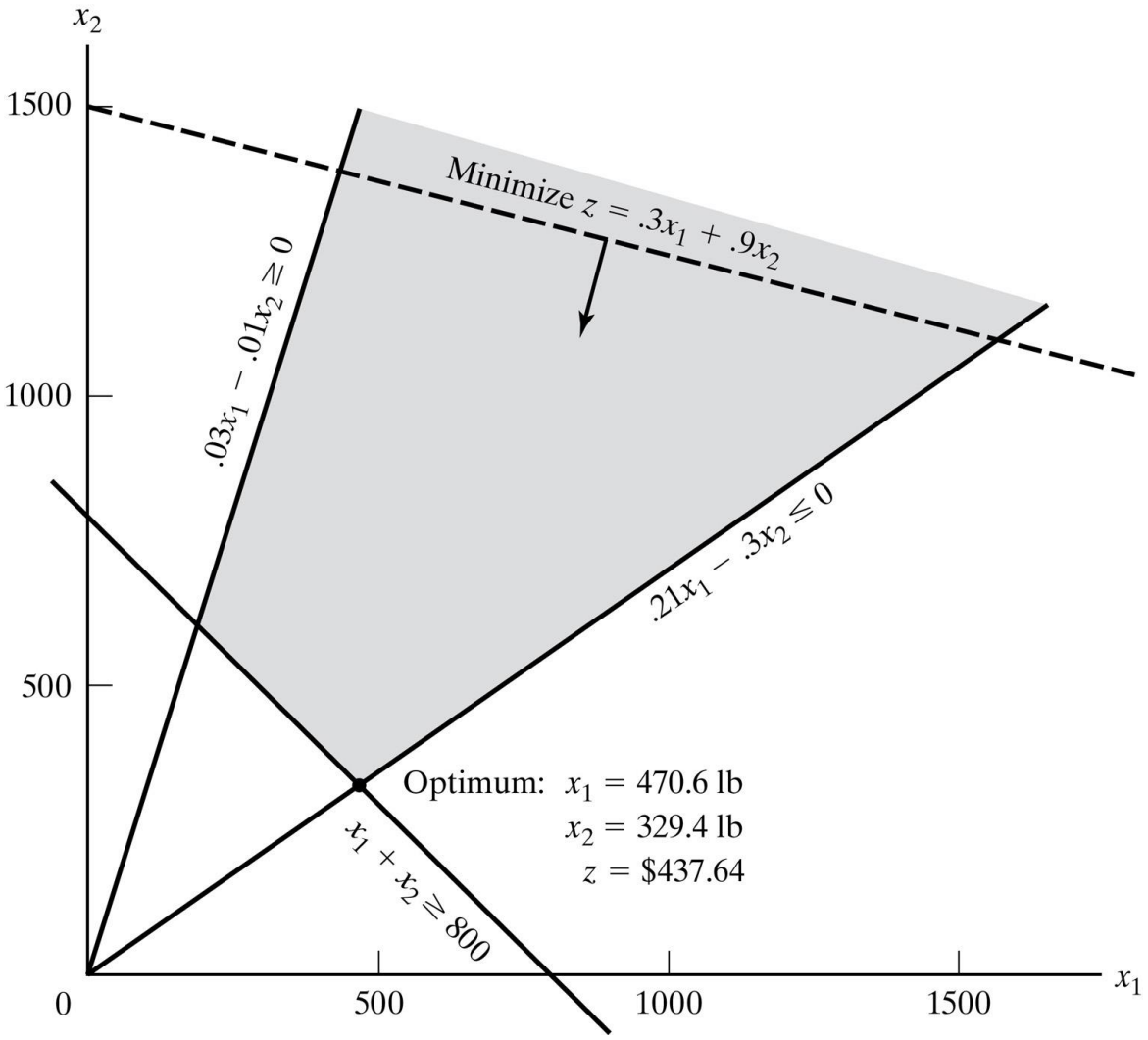
- What is the minimum cost feed mix?

x_1 =
x_2 =

minimize

s.t.

Min.
s.t.


x_2
1500
1000
500
0
500
1000
1500
x_1
Minimize $z = .3x_1 + .9x_2$
$.03x_1 - .01x_2 \geq 0$
$.21x_1 - .3x_2 \leq 0$
$x_1 + x_2 \geq 800$
Optimum: $x_1 = 470.6$ lb
$x_2 = 329.4$ lb
$z = \$437.64$

LP MODELING EXAMPLES

A Product Mix Problem:

- A TV company produces two types of models, A and B.
- There are two production lines, one for each model.
  - The capacity of the A production line is 70 sets per day,
  - The capacity of the B line is 50 sets per day.
- There are two departments P (picture tube) and C (chassis) both of which are used in the production of each model.
  - In department P, model A requires 1 labor hour and B requires 2 labor hours; a maximum of 120 labor hours can be assigned to department P.
  - In C, A requires 1 hour and B requires 1 hour; a maximum of 90 labor hours can be assigned to C.
- Profit contributions are $20 and $10 per set of A and B, respectively.
- If the company can sell as many sets as it can produce, what should be the daily production plan for each model?

Summary (as a table)

Decision variables:

Max.
s.t.

Important note:

A Blending Problem: The Green Lawn & Garden store is trying to sell its own brand of lawn fertilizer this year. It plans to sell two types of fertilizer, one high in nitrogen content and the other an all-purpose fertilizer. The fertilizers are mixed from two ingredients that contribute nitrogen and phosphorus in different amounts. The composition of each ingredient and cost per pound is given below.

Ingredient	Nitrogen, %	Phosphorus, %	Cost, $/lb
1	60	10	0.20
2	10	40	0.30

The season's demand is estimated to be 5,000 25-lb bags of high-nitrogen fertilizer and 7,000 25-lb bags of all-purpose fertilizer. The high-nitrogen fertilizer is to contain between 40 to 50% nitrogen and all-purpose fertilizer is to contain, at most, 20% phosphorus. How many pounds of each ingredient should the store purchase to satisfy the estimated demand at minimum cost?

Decision variables:

Objective Function:

Constraints:

The Complete LP-Model:

A Scheduling Problem: A personnel manager must schedule the security force in such a way as to satisfy the staffing requirements shown below:

Work time interval	1	2	3	4	5	6
Minimum # required	5	7	15	7	12	9

Staffing requirements time intervals are,
midnight-4am, 4am-8am, 8am-noon, noon-4pm, 4pm-8pm, and 8pm-midnight.

Officers work 8-hr shifts:
midnight-8am, 4am-noon, 8am-4pm, noon-8pm, 4pm-midnight, and 8pm-4am.

The manager wants to know how many officers should work each shift in order to minimize the total number of officers employed.

Decision variable:

Mathematical Model

A Transportation Model: A company has two plants and three warehouses. Plant P_1 can produce at most 100 units and Plant P_2 can produce at most 200 units of the same product. The sales potential at Warehouse W_1 is 150, 200 at Warehouse W_2, and 350 at Warehouse W_3. The sales revenue per unit at the warehouses are: \$12 in W_1, \$14 in W_2, and \$15 in W_3. The cost of manufacturing one unit at P_i and shipping it to W_j is given below:

Decision Variables:

Objective Function:

Complete LP Model:

A Multi-Period Inventory Model: A producer of polyurethane has a stock orders for the next n periods (e.g., 6 months); let d_i be the known demand in period i. Let c_i be the cost of producing a gallon during period i and K_i be the maximum amount that can be produced in period i (because of capacity limitation). Finally, let h_i denote the per unit cost of inventory in stock at the end of period i. Suppose that I_0 be the initial inventory at the beginning of period 1. Find a production inventory-holding plan that will satisfy the known demand schedule at minimum total cost.

Decision variable:

Define a state variable

I_i = gallons of inventory at the end of period i. So,

Complete LP Model:

Investment (Loan Policy Model)

Thriftem Bank is in the process of devising a loan policy that involves a maximum of $12 million. The following table provides the pertinent data about available types of loans.

Type of Loan	Interest Rate	Bad-debt ratio
Personal	.140	.10
Car	.130	.07
Home	.120	.03
Farm	.125	.05
Commercial	.100	.02

Bad debts are unrecoverable and produce no interest value.

Competitions with other financial institutions require that the bank allocate at least 40% of the funds to farm and commercial loans. To assist the housing industry in the region, home loans must equal 50% of the personal, car and home loans. The bank also has stated policy of not allowing the overall ratio of bad debts on all loans to exceed 4%.
Thriftem Bank wants to maximize its net return, the difference between interest revenue and lost bad debts.

Single Period Production Model

In preparation for the winter season, a clothing company is manufacturing parka and goose overcoats, insulated pants and gloves. All products are manufactured in four departments: cutting, insulating, sewing and packaging. The company has received firm orders for its products. The contract stipulates a penalty for undelivered items. The following table provides the pertinent data of situation.

	Time per units (hr)				
Department	**Parka**	**Goose**	**Pants**	**Gloves**	**Capacity (hr)**
Cutting	.30	.30	.25	.15	1000
Insulating	.25	.35	.30	.10	1000
Sewing	.45	.50	.40	.22	1000
Packaging	.15	.15	.10	.05	1000
Demand	800	750	600	500	
Unit Profit	$30	$40	$20	$10	
Unit Penalty	$15	$20	$10	$8	

Devise an optimal production plan for the company.

An Approach for Modeling

Step 1- Picture & Paraphrase: Initial understanding of the problem.

(i) Work in a flow-chart, input-output diagram, block diagram, process diagram, graph, schematic, etc. of the situation.

(ii) Then write a description of the problem without formula or data.

(iii) Show these to person(s) familiar with the problem for feedback & make sure that you are taking into account everything correctly.

Step 2 – Verbal Model: Break the problem into smaller elements.

(i) *Decisions* – something that can be evaluated or set to a value (how much to make, how many to hire/fire/promote, which warehouse to use, etc.)

(ii) *Constraints* – something that places limitation and/or requirement in the situation (budget, available rates of return, capacity, etc.). Constraints may fall in the following categories,

Inputs– what goes in the process? Are raw materials, capital, or labor restricted or limited in supply? Are there minimum amounts of input required?

Outputs– what is being produced? Are there sales requirements or limitations? Does a certain amount have to be kept in inventory?

Capacity– what capacity (time, labor, space) keeps you either from producing an infinite amount or from making more profit?

Ratio/Balance– what ratio between products or balance of ingredients must be maintained?

(iii) *Objective function* – what are you trying to optimize? May have many types of objective functions. How will it be measured (revenue, profit, market share, net present value, service level, etc.). Decision variables should directly or indirectly affect the objective. Should it be maximized or minimized?

Step 3 – Algebraic Model: transform verbal components of the model into algebraic representations.

Decisions– choose names or labels that describe each alternative, decide on their units.

Constraints– write each term as coefficient times the variables on the left-hand-side, an inequality ($\leq$ or $\geq$) or equality (=) sign, and a constant on the right-hand-side. Three types of constraints: Restrictions ($\leq$), Requirements ($\geq$) or Equalities (=).

Objective function– write as coefficients times decision variables (to be maximized or minimized).

Step 4: Computer Model: transfer the algebraic model to a spreadsheet (e.g. Excel) or other optimization software (e.g. TORA). Follow the spreadsheet/software input format. Document the model with comments. Once model has been solved, the results may have to be put in a more readable form (with additional comments, graphs, charts and/or written report of what has been discovered); needs validity and may need to perform what-if or sensitivity analysis.